



Quartz

△ Brazilian amethyst crystal, with broken base

Quartz is the third most common mineral on the Earth's crust, after ice and feldspar. Of all minerals, it has the greatest number of gem varieties, including prized gems such as amethyst, chalcedony, and agate. Quartz comes in two basic forms: crystalline (as distinct crystals), and cryptocrystalline (formed of microscopic crystalline particles). The optical and electrical properties of colourless, transparent quartz have led to its extensive use in lenses and prisms, and as oscillators for electronic devices such as watches.

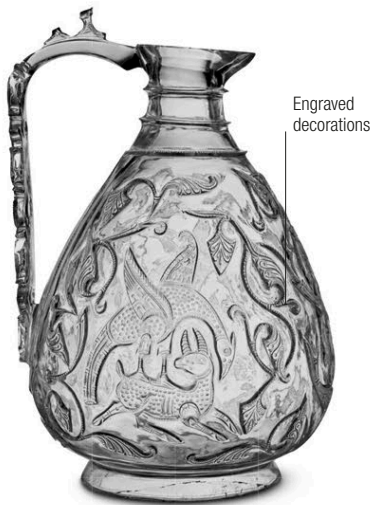
Stone of wonder

The word “quartz” comes from Old German and first appears in the writings of Georgius Agricola in 1530. However, long before this, Roman naturalist Pliny the Elder (23–79 CE) believed quartz to be ice that had been permanently frozen after great lengths of time, his evidence being that quartz is found near glaciers in the Alps, but not on volcanic mountains. Egg-sized, white quartz pebbles are found in Bronze-Age tombs in Europe, and in early Christian churches and chapels in Ireland and the North of England. Even now, rock crystal – transparent, colourless quartz – is commonly regarded in shamanistic practice as a “light-stone”, an instrument of clairvoyance between the visible and invisible. Australian Aborigines used rock crystal both as a talisman and to produce visions, while the Navajo believed that it first caused the Sun to cast its light upon the world.

Settings



Amuletic pendant | Set | This ancient amulet from the Egyptian New Kingdom takes the form of a lion's head carved from amethyst, set on a gold base featuring baboon figures. It dates from around 700 BCE.



Crystal ewer | Carved | This stunningly crafted ewer was carved and hollowed from a single piece of rock crystal in the Fatimid Period (969–1161 CE) in Egypt, the source of a number of rock crystal artefacts.



Duchess of Windsor's Cartier necklace | Set | Incorporating 29 smaller, step-cut amethysts and a large, central heart-shaped amethyst gem, this bib necklace is also set with turquoise and diamonds.

Specification

Chemical name Silicon dioxide | **Formula** SiO₂ | **Colours** All colours | **Structure** Trigonal | **Hardness** 7 | **SG** 2.65
RI 1.54–1.55 | **Lustre** Vitreous | **Streak** White



Cabochon



Mixed



Step



Pendeloque



Cameo



Locations
1 Brazil **2** Scotland **3** Spain **4** France **5** Swiss Alps
6 Russia **7** Sri Lanka **8** Madagascar